

WEEK 1&2

Question:

1. Sample Case study on class & objects CO1

Bank Account Management System:

Problem Statement: Design a system to manage bank accounts.

Classes: Account, Savings-Account, Current-Account.

Attributes: Account number, balance, account type, interest rate (for savings account), overdraft limit (for current account), etc.

Methods: Deposit, Withdraw, Calculate Interest, Transfer, etc.

Object Usage: Create objects for each customer's account and perform transactions like

deposits, withdrawals, and transfers.

Algorithm

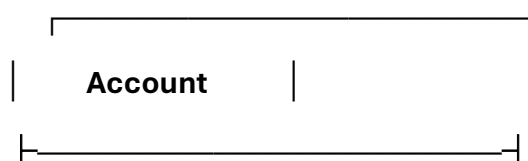
1. **Start** the program.
2. Create a Scanner object to take input from the user.
3. Create one **Savings Account** object with:
 - Account number
 - Initial balance
 - Interest rate
4. Create one **Current Account** object with:
 - Account number
 - Initial balance
 - Overdraft limit
5. Display the bank menu using an infinite while(true) loop.
6. Read the user's choice.
7. Use a switch statement to perform the selected operation:
 - **Deposit:** Select an account and deposit the given amount.

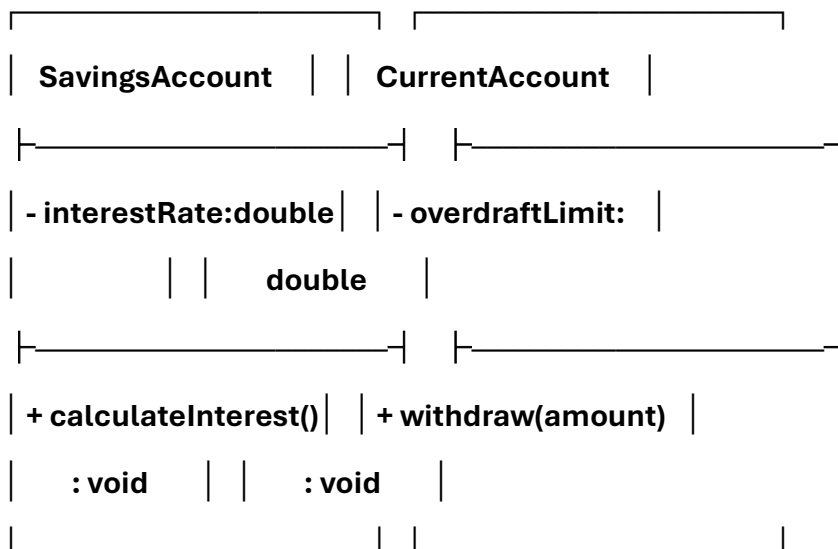
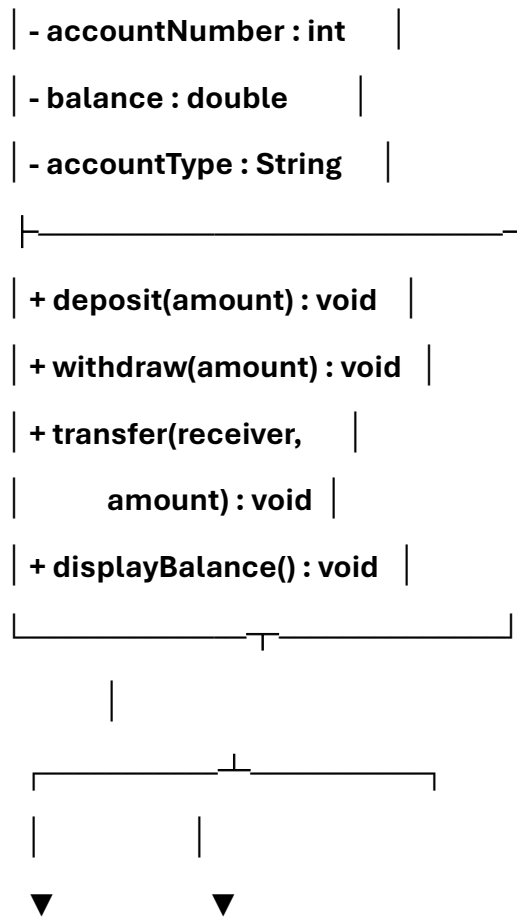
- **Withdraw:** Select an account and withdraw the given amount.
 - **Transfer:** Select the source and destination account and transfer the given amount.
 - **Savings Account Balance:** Display the savings account balance.
 - **Calculate Interest:** Calculate interest and add it to the savings account balance.
 - **Current Account Balance:** Display the current account balance.
 - **Exit:** Stop the program.
8. Repeat the menu until the user selects **Exit**.
 9. **Stop** the program.

Steps for Execution

1. Open **VS Code / Eclipse / any Java IDE**.
2. Create a new Java file named **BankAccount.java**.
3. Enter the complete Bank Account Management System code.
4. Save the file.
5. Compile the program.
6. Run the program.
7. The **Bank Menu** will be displayed.
8. Enter the required option.
9. Enter the account and amount whenever requested.
10. The selected transaction will be performed and the result will be displayed.
11. After the operation, the menu appears again because of the while(true) loop.
12. Continue performing transactions as required.
13. Select **7. Exit** to terminate the program.

UML class diagram:





Output:

EXPLORER

CSMD 247

Bankmanagement.java

PROBLEMS

OUTPUT

DEBUG CONSOLE

Bankmanagement

```
1. Deposit
2. Withdraw
3. Transfer
4. Savings Account Balance
5. Calculate Savings Interest
6. Current Account Balance
7. Exit
Enter your choice: 1
1. Savings Account
2. Current Account
Choose account: 2
Enter amount: 22
Amount deposited successfully.
Enter your choice: 6

--- Current Account ---
Account type:Current
Account Number: 102
Balance :5022.0
Enter your choice: 3
1. Savings ->Current
2. Current -> Savings
Choose transfer: 1
Enter amount: 211
Amount transferred successfully.
Enter your choice: 6

--- Current Account ---
Account type:Current
Account Number: 102
Balance :5233.0
Enter your choice: 
```

OUTLINE

TIMELINE

JAVA PROJECTS

0 0 Java: Ready